Rows Columns and Data Types

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Rows, Columns, and Data Types

> Prepare Rows Columns and Data Types: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Specify a table **schema**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-2/7, ai-4/4. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Debugging Strategies

Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

Look for: Design revealing tests for a function

Reference example, not the only acceptable wording: 4, 0 and 6.

Trace Tables

Without opening the old lesson, explain one key idea from Trace Tables. Give an example and a mistake that the idea helps you avoid.

Look for: Trace variable values before and after each loop iteration

Reference example, not the only acceptable wording: 2, 3 and 4; the stop value is excluded. The body runs three times.

Precision in Algorithms

Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

Look for: Translate a short paper calculation into a Python script

Reference example, not the only acceptable wording: packs = 4; per_pack = 6; total = packs * per_pack; print(total), each on its own line. The first two bind quantities, the third computes 24 and the fourth displays 24.

Assignment alone produces no displayed **output** in this script.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A row contains title='Sky', pages='120', available='false'. Explain why treating nonempty text as true can misread availability. Propose explicit conversions and a rejected-**input** case.

Concept to explain

Week 7: Rows, **Column**s, and **Data Type**s

Term	Meaning
row	One record or example in a table.
column	A table field containing the same kind of information across rows.
data type	A category of values, such as number, text or Boolean, with defined operations.
schema	A specification of required fields, types and structural rules for data.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

> Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach Week 13 as a two-session investigation using fictional data. A fictional classroom logs the duration of each activity.

Before comparing durations, define a **schema**: activity is nonempty text, minutes is an integer from 0 through 60, and each row describes one activity session. These decisions make data interpretable.

Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

Look up the value associated with the key "used" in that record.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can read a table **schema** and distinguish a numeric value from its text representation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Rows Columns and Data Types using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. **Schema**: title=text; pages=whole number; genre=category.

2. Bad record: pages='many' cannot be compared numerically.

3. Repair: verify the source, then enter 128—not a guess.

4. Consistent **type**s let sorting and calculations work predictably.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

Fields are Title, Pages and Borrowed; a row contains River, ninety and yes.

The teaching case

Assign intended **type**s and a repair.

Exact explanation to model

Why the result follows

Text, numeric and Boolean. Confirm ninety means 90, then convert and record the change; use a consistent Boolean representation.

Explain the added worked example

A fictional classroom logs the duration of each activity. Before comparing durations, define a **schema**: activity is nonempty text, minutes is an integer from 0 through 60, and each row describes one activity session.

These decisions make data interpretable.

Explain the added worked example · 1

The worked script prints True twice. **type**(value) is int checks the exact **type** here and avoids accepting a Boolean as a count. The chained comparison checks both boundaries.

The example assumes both keys exist; next week handles missing fields. A zero duration is allowed by this written **schema**.

Key terms

Term	Meaning
Schema	Rules for the fields and values in records
Type	A category such as integer or text
Unit	The quantity scale attached to a number
Constraint	A rule a valid record must satisfy

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

> **Would minutes of 61, "12" and 12 each pass? Explain the type distinction.**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Rows Columns and Data Types

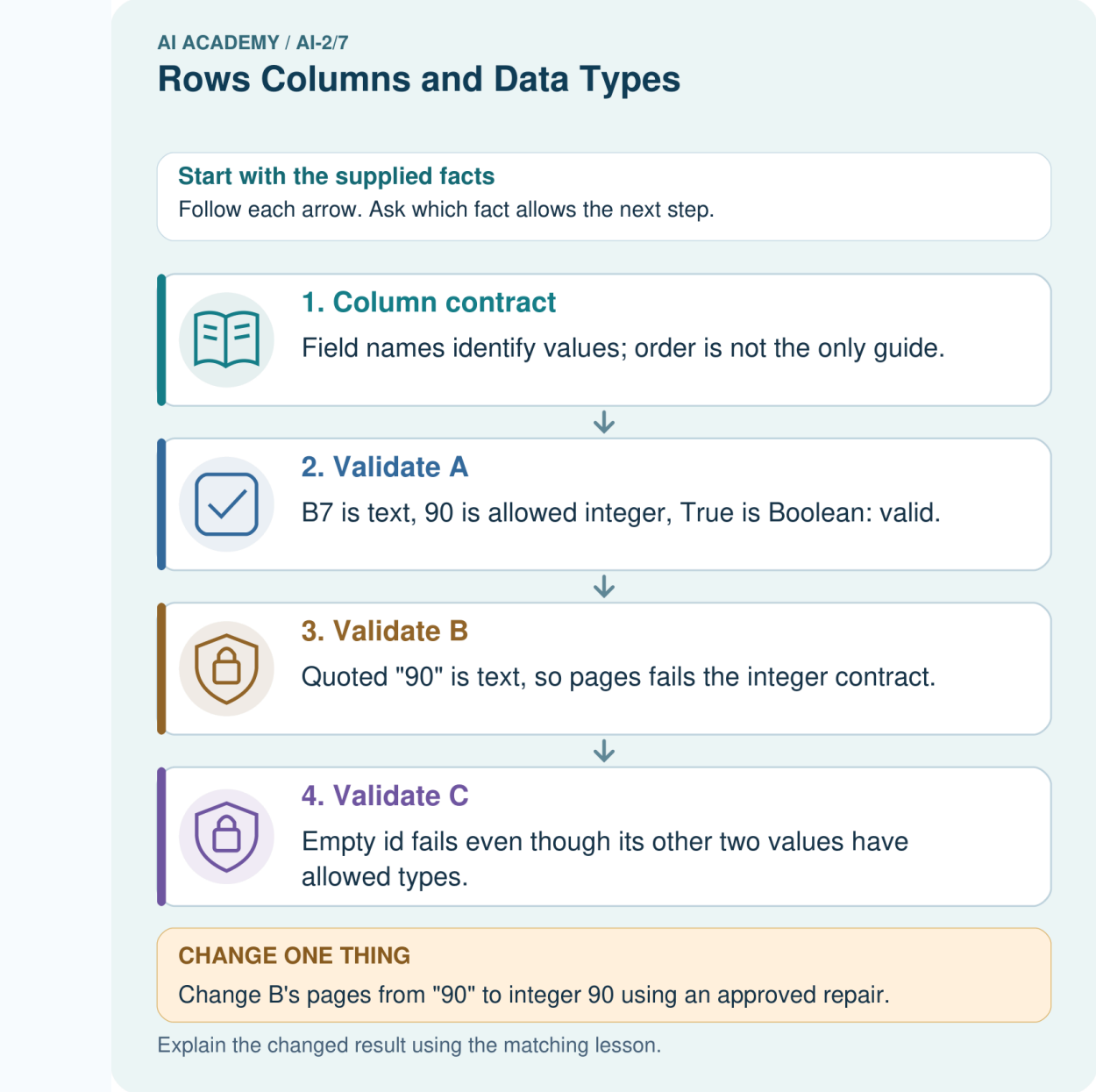
A fictional approved book catalogue has three fields per row.

The facts we will use

> **Supplied facts**
>
> **Schema**: id is nonempty text; pages is an integer 1–300; illustrated is Boolean True or False.

Supplied facts

Rows A={id:"B7",pages:90,illustrated:True}; B={id:"B8",pages:"90",illustrated:False}; C={id:"",pages:80,illustrated:True}.



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Column contract	Field names identify values; order is not the only guide.
2. Validate A	B7 is text, 90 is allowed integer, True is Boolean: valid.
3. Validate B	Quoted "90" is text, so pages fails the integer contract.
4. Validate C	Empty id fails even though its other two values have allowed types.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **schema** makes row meanings and expected **type**s explicit. The string "90" looks like a number when printed but does not meet this strict integer **schema**. A repair can be authorised later, but must not be assumed silently.

Missing or empty identifiers also prevent tracing an item back to its source.

What this example does not establish

Keep the stated limit

Schema validity does not confirm that the book really has 90 pages or that its metadata is accurate.

Change one thing in the pictured case

Change one thing

Change: Change B’s pages from "90" to integer 90 using an approved repair.

Predicted result

Prediction: B now satisfies the declared **schema**.

Why the result follows

Reason: Its id is nonempty and illustrated is Boolean; the page-**type** defect is removed.

Ask while showing the picture

Ask while showing the picture · 2

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Rows Columns and Data Types, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Fields are Title, Pages and Borrowed; a row contains River, ninety and yes.

2. State the decision to explain

Write the question to explain: Assign intended **type**s and a repair.

3. Reveal the worked reasoning

Reveal after **prediction**s: Text, numeric and Boolean. Confirm ninety means 90, then convert and record the change; use a consistent Boolean representation.

Hold back the independent task

Keep for an independent attempt: A table row is book_id B7, pages 84, available true. Name the **type** of each field and explain why pages stored as eighty-four complicates calculation.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write **unit**s and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

61 fails the range; "12" fails the integer **type**; 12 passes. Exit: No. The value has a permitted structure but still needs trustworthy collection **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.

Stage	Teacher action
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion practice

Completion task: Classify minutes values 0, 60, -1 and 61 under this **schema**.

Reasoning to listen for

Teacher check: 0 and 60 pass; -1 and 61 fail.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Rows **column**s and **schema**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: A row can satisfy every **schema** rule and still be a wrong **observation**.

Schema validation checks our stated structure and **constraint**s, not whether someone measured honestly or selected a representative sample.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

> Which two rows originally fail, and do they fail for the same reason?

Reasoned teacher answer

B fails pages **type**; C fails empty id. A passes all three field checks.

Changed-case reasoning: B now satisfies the declared **schema**. Its id is nonempty and illustrated is Boolean; the page-**type** defect is removed.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Rows Columns and Data Types investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Audit a 10-row book table. Circle mixed **type**s, missing values, duplicated rows, and fields with unclear names; document each repair.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A successful run proves the conclusion: A row can satisfy every **schema** rule and still be a wrong **observation**.

Schema validation checks our stated structure and **constraint**s, not whether someone measured honestly or selected a representative sample.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: define a tray label and integer count 0 through 20. Require tests at 0, 20, -1 and 21 and a text count.

A convenient shortcut preserves meaning: If one row records 120 seconds in the minutes field, a simple numeric total becomes misleading.

Correct the source or explicitly convert and document it; renaming a **column** alone does not convert values.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Rows Columns and Data Types questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Text, numeric and Boolean. Confirm ninety means 90, then convert and record the change; use a consistent Boolean representation.

Independent question

Independent question · withhold the key

A table row is book_id B7, pages 84, available true. Name the **type** of each field and explain why pages stored as eighty-four complicates calculation.

Expected independent answer

Teacher key · Expected independent answer

book_id is text, pages is a number, and available is Boolean. Text words need conversion before numeric arithmetic or comparison.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions

Criterion	0 points	1 point	2 points
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “All table entries behave as numbers.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Parse pages as a number after validation. Map only agreed strings such as 'true' and 'false' to Booleans; reject 'unknown' or represent it explicitly as missing. Do not rely on nonempty-string truthiness.

Type consistency does not prove that the catalogue's facts are correct.

Explanations for the additional workbook examples

W1 State a schema

Workbook question · W1

List the required fields, **type**s and duration boundaries for the worked table.

Teacher reasoning · W1

activity: nonempty text; minutes: integer from 0 to 60 inclusive.

Teacher reasoning · W1

Check both fields. activity must exist as nonempty text. minutes must be an exact integer and satisfy 0 <= minutes <= 60. Inclusive means equality is accepted at both ends.

Structure, **type** and range are separate checks; passing them does not prove the duration was measured accurately.

W2 Classify records

Workbook question · W2

Classify minutes values 0, 60, -1 and 61 under this **schema**.

Teacher reasoning · W2

0 and 60 pass; -1 and 61 fail.

Teacher reasoning · W2

Apply the inclusive range to each integer. Both 0 and 60 are on allowed boundaries. -1 is below the minimum and 61 is above the maximum. Require the reason for each rejection, not only pass/fail labels.

W3 Repair a unit problem

Workbook question · W3

A source says 180 seconds but the row says minutes: 180. Give a documented correction.

Teacher reasoning · W3

Divide by 60 to record 3 minutes and retain the source value and conversion note.

W4 Design a schema independently

Workbook question · W4

For a fictional shelf audit, define label and book_count fields. Require a nonempty text label and an integer count from 0 to 40. Give two valid and two invalid records with reasons.

Teacher reasoning · W4

Examples: A/0 and B/40 valid; empty label/5 invalid label; C/41 invalid range. Text "5" also fails the count **type**. Accept alternatives meeting the **written rule**s.

W5 Separate validity and truth

Workbook question · W5

A row says Reading, 12 and passes. Does that establish the session really lasted **12 minutes**?

Teacher reasoning · W5

No. The value has a permitted structure but still needs trustworthy collection **evidence**.

Assess independent transfer

Independent question · withhold the key

Selected case: For a fictional shelf audit, define label and book_count fields. Require a nonempty text label and an integer count from 0 to 40. Give two valid and two invalid records with reasons.

Selected case answer

Teacher key · Selected case answer

Examples: A/0 and B/40 valid; empty label/5 invalid label; C/41 invalid range. Text "5" also fails the count **type**. Accept alternatives meeting the **written rule**s.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Rows Columns and Data Types to Transparent Book Recommender, adapt support and plan a later changed-case check.

Transparent Book Recommender

Store one book per row and keep reader preferences in a separate record.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled **column**s. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 8: Operational Labels. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use a **schema** card beside each record. Extension: explain how allowing decimal durations would change both **type** and rounding rules.

Independent W4 **evidence**: Examples: A/0 and B/40 valid; empty label/5 invalid label; C/41 invalid range. Text "5" also fails the count **type**. Accept alternatives meeting the **written rule**s.

Record whether the learner needed no help, a hint or explanation. Fresh retry: define a tray label and integer count 0 through 20. Require tests at 0, 20, -1 and 21 and a text count.

Next week: Missing values and validation. Reuse this week's explicit contracts and **evidence**.

Plan the delayed return

Suggested return points in this level: ai-2/8, ai-2/10, ai-2/13. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Publish the recommender's row **schema** and validation results before ranking books, preserving repair provenance.

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Prepare, teach and assess

> Use this preparation summary alongside the detailed Rows Columns and Data Types lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Specify a table **schema**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
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> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Classify minutes values 0, 60, -1 and 61 under this **schema**.

Supported practice answer

Reasoned practice answer: 0 and 60 pass; -1 and 61 fail.

Check every learner before discussion: Use the worked case for Rows **column**s and **schema**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: A row can satisfy every **schema** rule and still be a wrong **observation**.

Schema validation checks our stated structure and **constraint**s, not whether someone measured honestly or selected a representative sample.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the

reasoning goal.

Extend the reasoning

Extension: Change B's pages from "90" to integer 90 using an approved repair. Require a **prediction** and a reason before comparing. Expected result: B now satisfies the declared **schema**.

Reason: Its id is nonempty and illustrated is Boolean; the page-**type** defect is removed.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

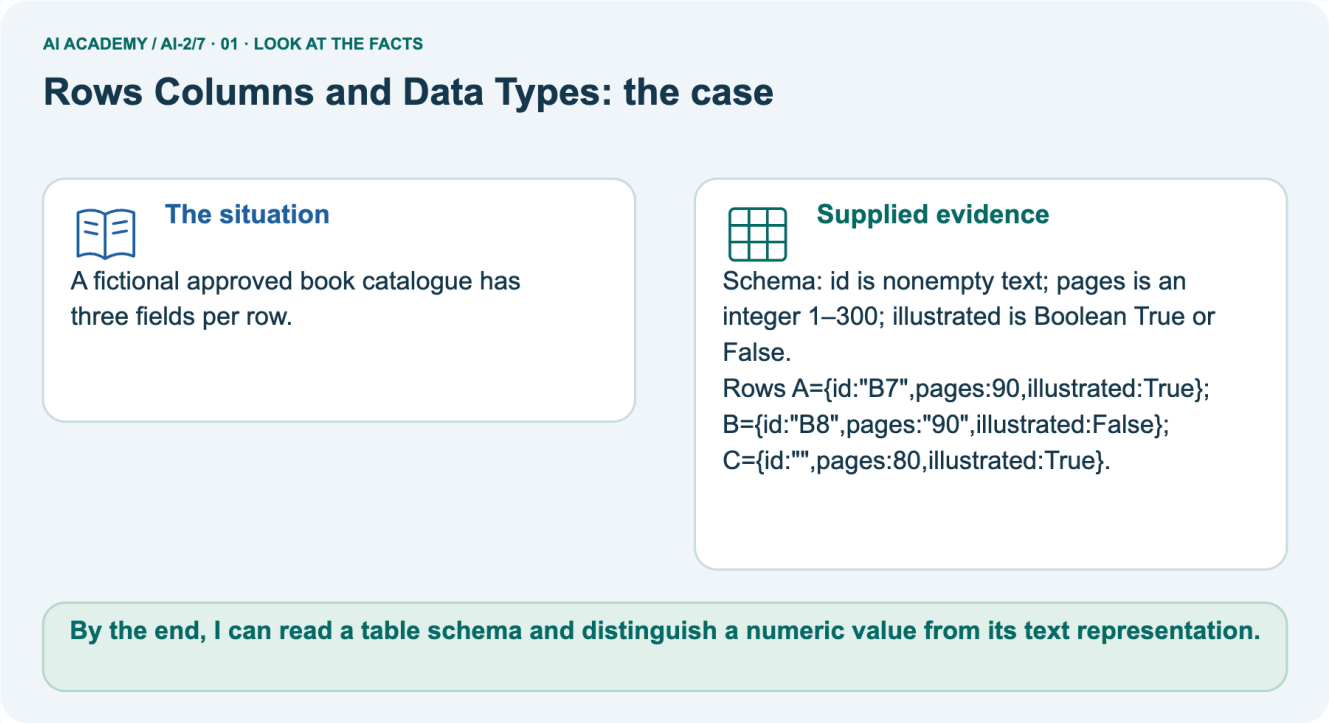
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Visual concept journey

> Show one Rows Columns and Data Types graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Rows Columns and Data Types: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional approved book catalogue has three fields per row. Supplied facts: **Schema**: id is nonempty text; pages is an integer 1–300; illustrated is Boolean True or False.

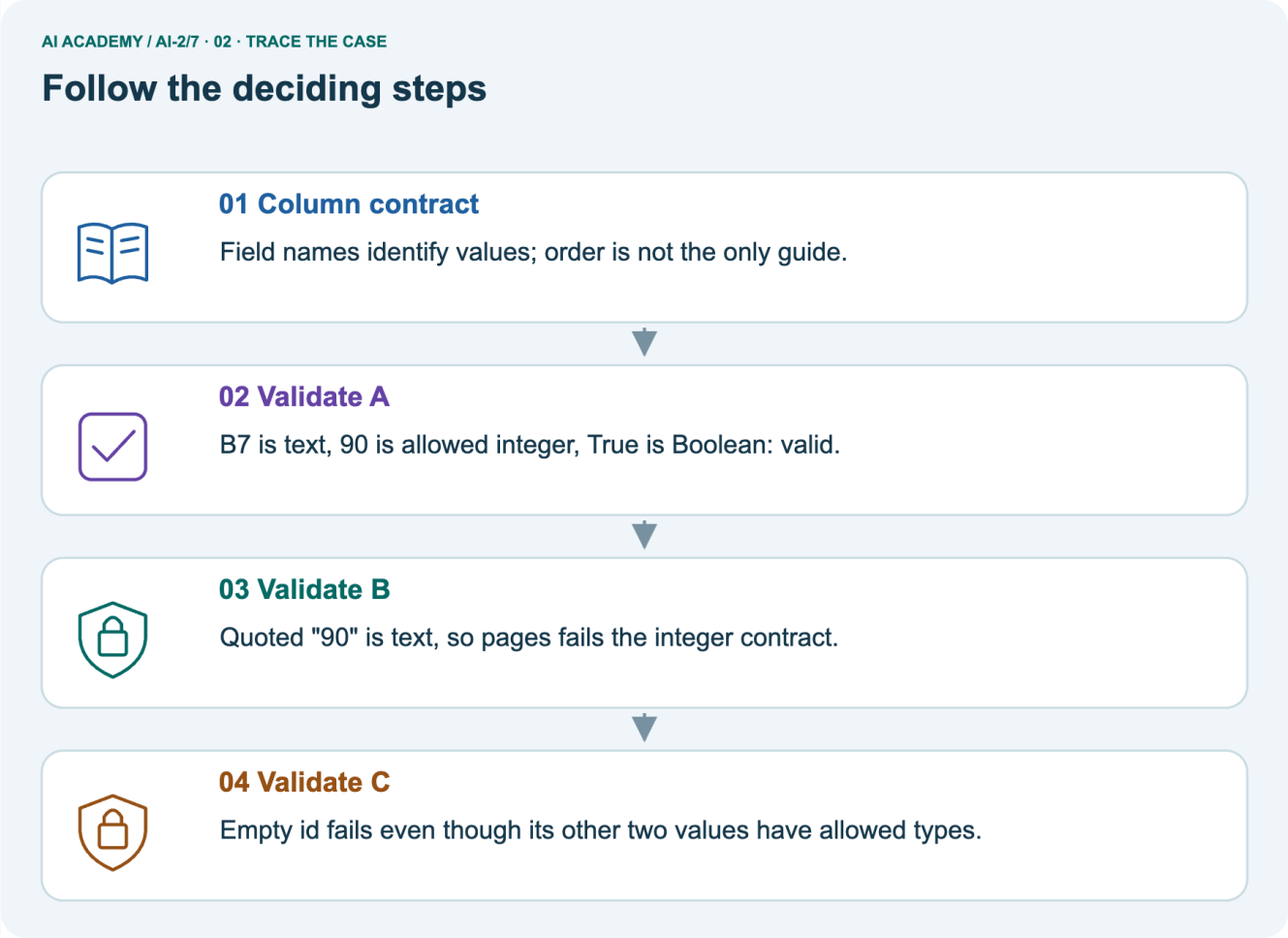
Rows A={id:"B7",pages:90,illustrated:True}; B={id:"B8",pages:"90",illustrated:False}; C={id:"",pages:80,illustrated:True}.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

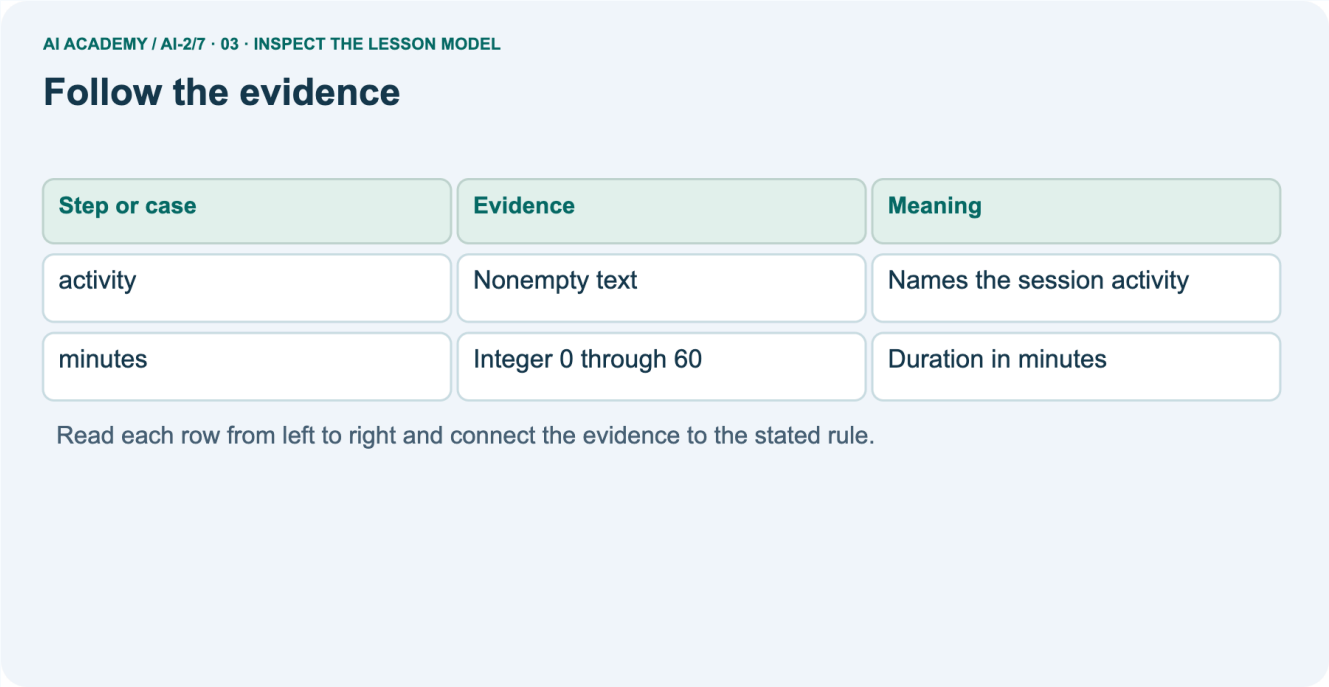
Column contract: Field names identify values; order is not the only guide. → Validate A: B7 is text, 90 is allowed integer, True is Boolean: valid. → Validate B: Quoted "90" is text, so pages fails the integer contract. → Validate C: Empty id fails even though its other two values have allowed **type**s.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Follow the evidence



Follow the labelled picture with the matching explanation. Open the vector diagram

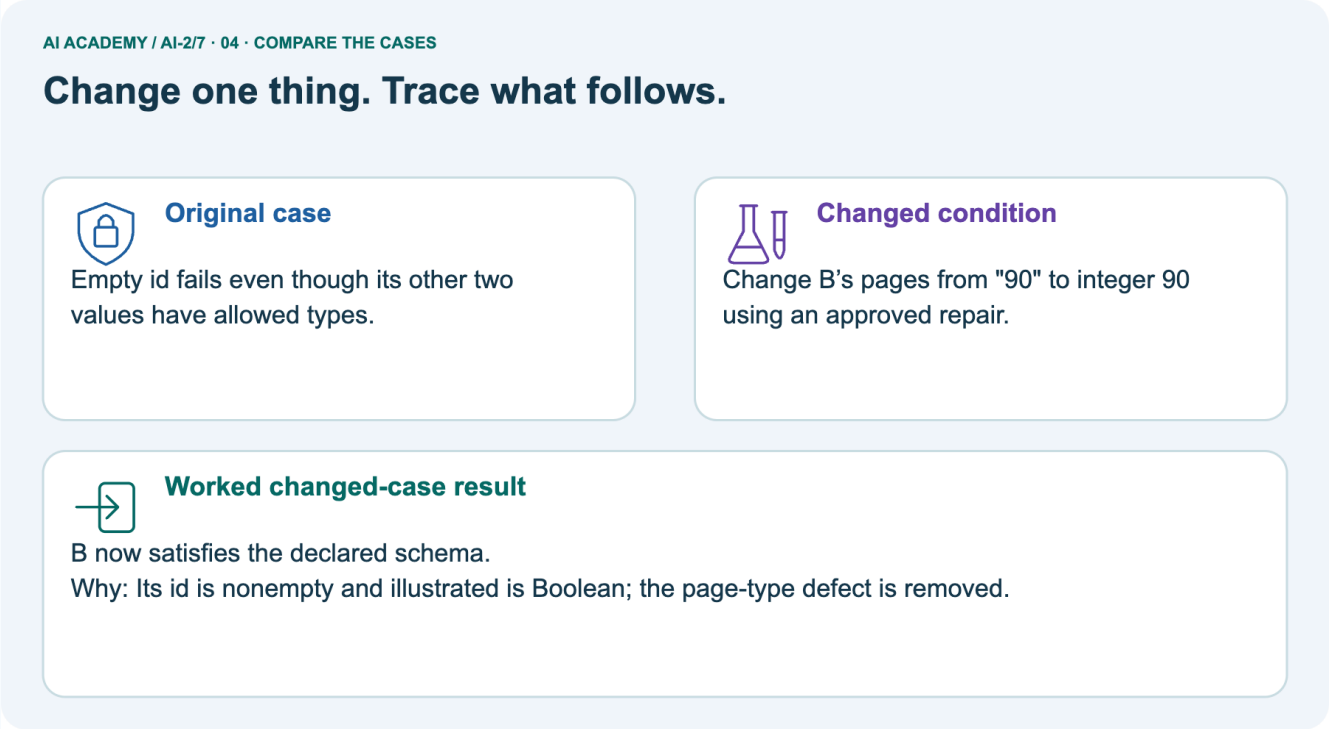
Follow the **evidence**. activity | Nonempty text | Names the session activity. minutes | Integer 0 through 60 | Duration in minutes

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Empty id fails even though its other two values have allowed **type**s. Changed condition: Change B's pages from "90" to integer 90 using an approved repair. Expected result: B now satisfies the declared **schema**.

Reason: Its id is nonempty and illustrated is Boolean; the page-**type** defect is removed.

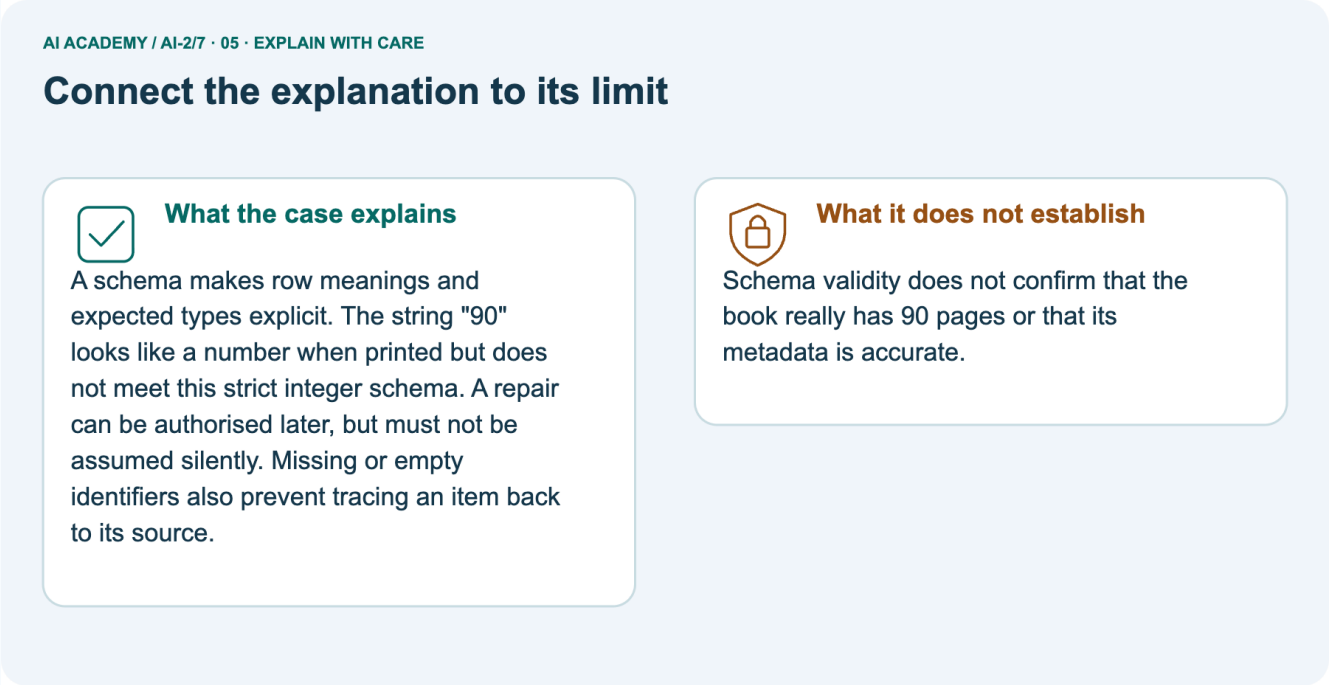
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Its id is nonempty and illustrated is Boolean; the page-**type** defect is removed.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **schema** makes row meanings and expected **type**s explicit. The string "90" looks like a number when printed but does not meet this strict integer **schema**. A repair can be authorised later, but must not be assumed silently.

Missing or empty identifiers also prevent tracing an item back to its source. Limit: **Schema** validity does not confirm that the book really has 90 pages or that its metadata is accurate.

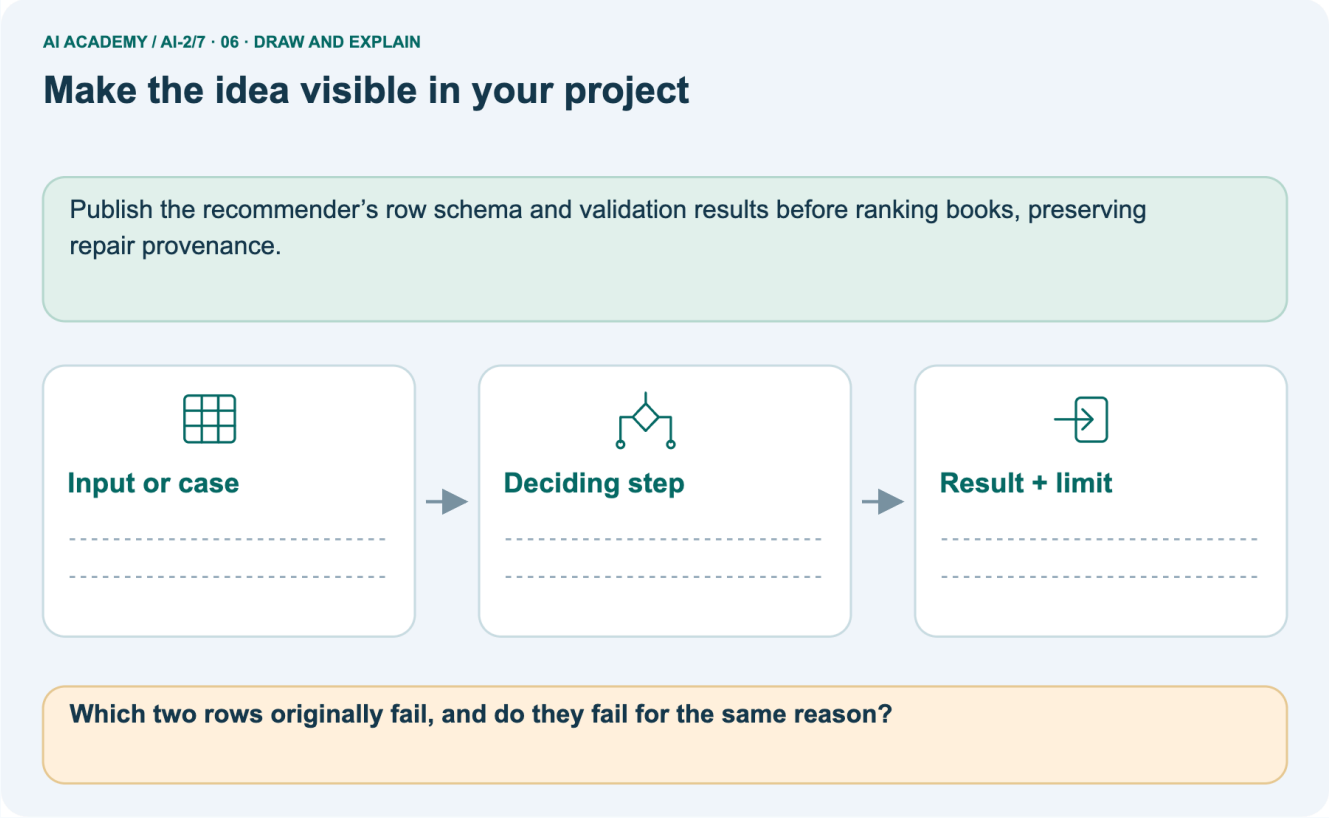
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: **Schema** validity does not confirm that the book really has 90 pages or that its metadata is accurate.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Publish the recommender’s row **schema** and validation results before ranking books, preserving repair provenance. Draw your **input**, deciding step, result and limit.

Supported question: Which two rows originally fail, and do they fail for the same reason?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: B fails pages **type**; C fails empty id. A passes all three field checks.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.